

# Maths

Unit 4 - Addition and  
Subtraction

**WALT:**

**6Nc.02** - Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

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**With support, I  
can answer  
questions that use  
letters to  
represent  
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**Independently, I  
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**I can specialise by  
showing all the  
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## Retrieval Practice

|                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Last Lesson</p> <p>Mr. Rhys says:<br/>"6 - y is the same as y - 6".<br/>Is he correct? Explain your answer.</p>                                   | <p>Last Week</p> <p>Ms. Colette has 500k VND in her bank account.<br/>She buys a bracelet for 875k VND.<br/>How much money does she have in her bank account?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Last Unit</p> <p>Ms. Alice created a sequence:<br/><br/>4, 8, 12, 16, 20, 24<br/>She says, "The position to term rule is +4." Is she correct?</p> | <p>Last Year</p> <p>8 What are the possible values of  and  when<br/> +  +  +  = 1.3 kg?</p> |

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**Pg. 58 - 61**

- 1 Cheng plays a board game using a dice.

He uses the instructions together with his dice score to work out how many spaces he moves.

$d$  represents the dice score.

For example:

| Score                                                                             | Calculate | Spaces moved |
|-----------------------------------------------------------------------------------|-----------|--------------|
|  |           |              |

Work out how many

|   | Score                                                                              |
|---|------------------------------------------------------------------------------------|
| a |   |
| b |   |
| c |  |

- 4 Martha buys 2 more pairs of socks than shoes.



- a Copy and complete the table where  $x$  represents the number of pairs of shoes and  $y$  represents the number of pairs of socks.

|     |   |   |   |
|-----|---|---|---|
| $x$ | 1 | 2 |   |
| $y$ | 3 |   | 6 |

- b Write a number sentence linking  $x$ ,  $y$  and 2.

- 5 This puzzle has 9 pieces.



Hassan places 1 or more shapes and Sanjay places the other shapes.

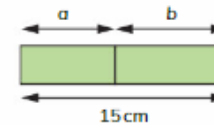
- a Copy and complete this table to show the number of pieces each person places.

|                                         |  |  |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|--|--|
| $x$ (number of pieces placed by Hassan) |  |  |  |  |  |  |  |  |  |
| $y$ (number of pieces placed by Sanjay) |  |  |  |  |  |  |  |  |  |

- 6  $a$  and  $b$  represent the lengths, in centimetres, of two strips of card.  $b$  is 3 cm longer than  $a$ .

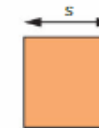
The two strips are placed end to end.

The total length is 15 cm.



Find the values of  $a$  and  $b$ .

- 7 The perimeter ( $p$ ) of a square is the sum of the lengths of the sides.  $s$  represents the length, in centimetres, of a side.



$$p = s + s + s + s$$

- a What is the value of  $p$  when  $s = 5$ ?  
 b What is the value of  $p$  when  $s = 7$ ?  
 c What is the value of  $s$  when  $p = 32$ ?
- 8 This isosceles triangle has a perimeter of 15 cm.

$x$  and  $y$  represent the lengths, in centimetres, of the sides of the triangle shown in the diagram.

### Think like a mathematician

$a$ ,  $b$  and  $c$  each represent a whole number from 1 upwards.

$$a + b + c = 7$$

Find all the possible values for  $a$ ,  $b$  and  $c$ .  
 How many different solutions can you find?

You will show you are specialising when you find all the possible values for  $a$ ,  $b$  and  $c$ .

- a Find three possible sets of  $a$ ,  $b$  and  $c$ .  
 b Write a formula for the perimeter of the triangle.

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1 Cheng plays a board game using a dice.

He uses the instructions together with his dice score to work out how many spaces he moves.

$d$  represents the dice score.

For example:

| Score                                                                             | Calculate | Spaces moved |
|-----------------------------------------------------------------------------------|-----------|--------------|
|  | $d + 4$   | 9 spaces     |

Work out how many spaces Cheng moves.

|   | Score                                                                               | Calculate | Spaces moved |
|---|-------------------------------------------------------------------------------------|-----------|--------------|
| a |   | $d - 3$   | <b>3</b>     |
| b |  | $6 - d$   | <b>4</b>     |
| c |  | $4 + d$   | <b>7</b>     |

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**2** For each pair of expressions write 'equal' or 'not equal'.

**a**  $d + 4$        $4 + d$       **equal**

**b**  $2 + d$        $d + 2$       **equal**

**c**  $5 - d$        $d - 5$       **not equal**

**3** Khalid says, ' $d + 3$  is the same as  $3 + d$  so  $d - 3$  must be the same as  $3 - d$ .' Is Khalid correct? Explain your answer.

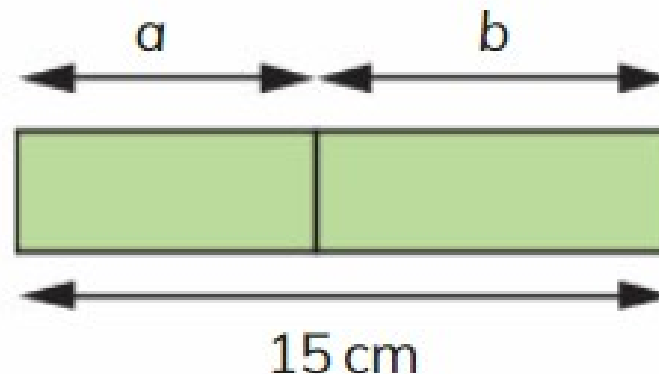
**Khalid is incorrect. Due to the commutative law it does not matter which number is written first in addition questions but it does in subtraction questions.**

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- 6  $a$  and  $b$  represent the lengths, in centimetres, of two strips of card.  
 $b$  is 3 cm longer than  $a$ .

The two strips are placed end to end.

The total length is 15 cm.



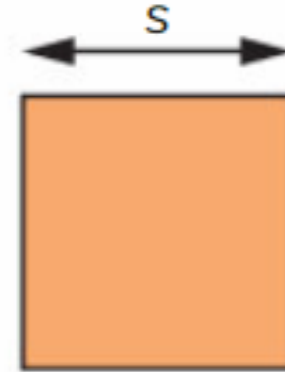
$$\begin{aligned} a &= 6\text{cm} \\ b &= 9\text{cm} \end{aligned}$$

Find the values of  $a$  and  $b$ .

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**7** The perimeter ( $p$ ) of a square is the sum of the lengths of the sides.

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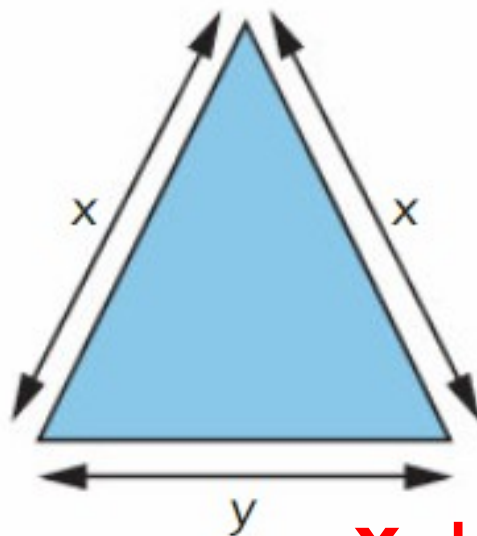
$$p = s + s + s + s$$

- a** What is the value of  $p$  when  $s = 5$ ? **20**
- b** What is the value of  $p$  when  $s = 7$ ? **28**
- c** What is the value of  $s$  when  $p = 32$ ? **8**

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8 This isosceles triangle has a perimeter of 15 cm.

$x$  and  $y$  represent the lengths, in centimetres, of the sides of the triangle shown in the diagram.



$$x + x + y = 15$$

- a Find three possible sets of values for  $x$  and  $y$ .
- b Write a formula for the perimeter ( $p$ ) of the triangle using  $x$  and  $y$ .

$$Y = 13$$

$$X = 1$$

$$Y = 11$$

$$X = 2$$

$$Y = 9$$

$$X = 3$$

$$Y = 7$$

$$X = 4$$

$$Y = 3$$

$$X = 6$$

$$Y = 1$$

$$X = 7$$

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## **Homework**

**Maths - Workbook - Pages 50 - 54**

**ESL - Workbook pages 20 - 23**

**Due - Thursday 24<sup>th</sup> September**

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## Plenary

**If  $5b = 20$  what does  $b =$  ?**

**$b = 4$**

**Why do you think we don't use the multiplication sign "x" ?**